

BCA 3rd SEMESTER EXAM. 2014
OBJECT-ORIENTED PROGRAMMING USING C++ CODE- 303301

Time: 3 hours

Full Marks: 60

Instructions:

- i. The Marks are indicated in the right -hand margin.
- ii. There are **SEVEN** questions in this paper.
- iii. Attempts **FIVE** question in all.
- iv. Question Nos. **1** and **2** are compulsory.

1. Answer the following as directed (any six).

6*2=12

(a) Base class pointer cannot point to derived class

(State True or False)

(b) Which of the following is not the member of class

(I) Static function

(II) Friend function

(III) Const function

(IV) Virtual function

(c) Which of the following access specifier is used in a class definition by default?

(I) Protected

(II) Public

(III) Private

(IV) Friend

(d) What happens when a class with parameterized constructor and having no default Constructor is used in a program and we create an object that needs a zero-arguments constructor?

(e) If the programmer does not explicitly provide a destructor, then ____ creates an empty destructor.

(Fill In the blanks)

(f) What is the use of function call operator?

(I) Overloading the method

(II) Overloading the objects

(III) Overloading the parameters

(IV) none of the above

(g) Which of the following advantages we lose by using multiple inheritances?

(I) Dynamic binding

(II) Polymorphism

(III) Both (i) and (ii)

(IV) none of the above

akubihar.com (h) Which of the following is referred by pointers to member?

(I) Static members of class objects

(II) Non-static members of class objects

(III) Referring to whole class

(IV) none of the above

(i) What does derived class not inherit from the base class?

(I) Friends

(II) Operator = () members

(III) Constructor and destructor

(IV) A of the above

(j) Which other keyword are also used to declare the class other than class?

(I) Struct

(II) Union

(III) Object

(IV) both (i) and (ii)

2. Answer any three of the following:**3*4=12**

(a) What is an operator function? Describe its syntax.

(b) Explain the characters of constructor and destructor.

(c) Write syntax for overloading extraction and insertion operator with friend function.

(d) Explain the mechanism for accessing private member of the base class using pointers.

(e) Describe various errors trapping function.

Answer any three of the following:**3*12=36****3. (a) What is function overloading? What are the rules for defining overloaded functions? Precautions**

should us tae while overdoing function?

6

(b) Write a program in C++ to use bit fids with cases display the contents of the objects.

6

4. (a) How are the constructor and destructor called explicitly? What is the difference between their calling methods?

6

(b) Write a program in C++ to pass an object to constructor and carry out copy constructor

display contents of the objects.

6

5. What is operator overloading? Explain the conversion of basic data type to class type and class type to basic type with appropriate example.

12

6. (a) Describe various type of inheritance with example.

6

(b) Write a program in C++ to declare class X Y and Z. Each class contains one character array as

a data member. Apply multiple inheritance. Concatenate strings of class X and Y and store it in Z. Show all the three strings. Use constructor and destructor.

6

7. (a) What are templates? Explain its need. How can normal function be declared as template function?

(b) Write a program in C++ to use exception handing with constructor and destructor.

6*2=12

Code : 303301

B.C.A. 3rd Semester Theory Examination, 2017

Object Oriented Programming using C++

Time : 3 Hrs

Full Marks : 60

Instructions :

- (i) The questions are of equal value.
- (ii) There are Seven questions in this Paper.
- (iii) Attempt Five questions in all.
- (iv) Question Nos. 1 & 2 are compulsory.

Answer any six Questions:

2×6=12

- (a) Give the evolution diagram of OOPS concept.
- (b) Write any four features of OOPS.
- (c) What is constructor?
- (d) What is an inline Function?
- (e) What are the blocks used in the Exception Handling?
- (f) Write the syntax of function template.
- (g) What is meant by inheritance?
- (h) What are pure virtual functions?
- (i) What are streams?
- (j) What are the functions that the file stream class provides?

Answer any three of the following:

4×3=12

- (1) What is a scope resolution operator? Give example.

P.T.O.

- (b) Briefly explain copy constructor.
- (c) Discuss the need for exception with try, catch and throw keywords.
- (d) What is polymorphism? Briefly explain the different types of polymorphism.
- (e) What are the types of File access? Write an example program for sequential access.

Answer any three questions:

12×3=36

- 3. Explain object-oriented paradigm with all its essential elements.
- 4. What is operator overloading? How many arguments are required in the definition of an overloaded binary operator?
- 5. What is class template? Write the syntax for class template. Write an example program for class template.
- 6. Explain abstract class with suitable C++ coding.
- 7. What is meant by inheritance? Explain its types with an example.

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Code : 303301

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BCA 3rd Semester Exam., 2018

**OBJECT-ORIENTED PROGRAMMING
USING C++**

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. **1** and **2** are compulsory.

1. Choose the correct option of the following
(any six) : 2×6=12

(a) Adding a derived class to a base class requires fundamental changes to the base class.

- (i) True
- (ii) False

(b) To expose a data member to the program, you must declare the data member in the ____ section of the class.

- (i) common
- (ii) exposed
- (iii) public
- (iv) unrestricted
- (v) user

(Turn Over)

(c) Hiding the complexity is known as

- (i) abstraction
- (ii) encapsulation
- (iii) data hiding
- (iv) composition

(d) Which of the following is a mechanism by which object acquires the properties of another object?

- (i) Encapsulation
- (ii) Abstraction
- (iii) Inheritance
- (iv) Polymorphism

(e) If I want to have common functions in a class and want to defer implementations of some other functions to derived classes, then we need to use

- (i) an interface
- (ii) an abstract class
- (iii) a friend class
- (iv) a static class

(3)

(f) Can main() function be made private?

- (i) Yes, always
- (ii) Yes, if program doesn't contain any classes
- (iii) No, because main function is user defined
- (iv) No, never

(g) cout stands for

- (i) class output
- (ii) character output
- (iii) console output
- (iv) call output

(h) Which of the following cannot be declared static? <https://www.akubihar.com>

- (i) Class
- (ii) Object
- (iii) Functions
- (iv) Both (i) and (ii)

(4)

(i) The generic type in a template function

- (i) must be T
- (ii) can be T
- (iii) cannot be T for functions you create, but may be for C++'s built-in functions
- (iv) cannot be T

(j) A function that is called automatically each time an object is destroyed is a

- (i) constructor
- (ii) destructor
- (iii) destroyer
- (iv) terminator

2. Answer any *three* of the following questions:

- (a) Define the characteristics of OOP.
- (b) Write the syntax for declaration of overloading extraction and insertion operator with friend function.

- (c) Explain the visibility of base class members for the access specifiers :
private, protected and public while creating the derived class.
- (d) How to create and destroy objects dynamically?
- (e) Explain the relationship between base class and derived class.

- 3. Explain the differences between class and structure with example. 12
- 4. What is operator overloading? What are the restrictions on operator overloading? Write C++ program to overload addition (+) operator for adding two complex numbers. 12
- 5. List the characteristics of a constructor. Write a C++ program to display distance. Define a suitable parameterized constructor with default values for the class 'distance' with data members 'feet' and 'inches'. 12
- 6. What is abstract class? Explain the different types of inheritance in C++. 12
- 7. Define exception handling. Explain the use of try, catch and throw for exception handling in C++ with example. 12

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(2)

BCA 3rd Semester Exam., 2019

**OBJECT-ORIENTED PROGRAMMING
USING C++**

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Choose the correct answer (any six) : 2×6=12

(a) Unsigned integers in C++ are

- (i) negative numbers
- (ii) non-negative numbers
- (iii) float numbers
- (iv) None of the above

20AK/413

(Turn Over)

(b) Which of the following correctly declares an array?

- (i) `int array[10];`
- (ii) `int array;`
- (iii) `array{10};`
- (iv) `array array[10];`

(c) How many types of constructor are there in C++?

- (i) 1
- (ii) 2
- (iii) 3
- (iv) 4

(d) Pick out the correct statement about string template.

- (i) It is used to replace a string.
- (ii) It is used to replace a string with another string at runtime.
- (iii) It is used to delete a string.
- (iv) None of the above

20AK/413

(3)

(e) Which is used to define the member of a class externally?

- (i) :
- (ii) ::
- (iii) #
- (iv) None of the above

(f) Which is more efficient while calling the functions?

- (i) Call by value
- (ii) Call by reference
- (iii) Call by pointer
- (iv) None of the above

(g) Which of the following is used to implement the C++ interfaces?

- (i) Absolute variables
- (ii) Abstract classes
- (iii) Constant variables
- (iv) None of the above

(Turn Over)

(4)

(h) What is the purpose of abstract class?

- (i) To provide help with database connectivity
- (ii) To provide data input to other classes
- (iii) To provide security to other classes
- (iv) To provide an appropriate base class from which other classes can inherit

(i) What is default visibility mode for members of classes in C++?

- (i) Private
- (ii) Public
- (iii) Protected
- (iv) Depends

(j) Identify the correct statement

- (i) Namespace is used to group class, objects and functions.
- (ii) Namespace is used to mark the beginning of the program.
- (iii) Namespace is used to separate the class, objects
- (iv) None of the above

(Continued)

2. Answer any three of the following : 4×3=12

- (a) Explain the basic data types and type modifiers in C++.
- (b) What are static data member? What are their benefits? Give example.
- (c) What is abstract class? What is its benefit?
- (d) What are the different types of operators in C++? What do you understand by precedence of operator?
- (e) What is a reference variable? Explain how to create and use reference variable with example.

3. Why is C++ called object-oriented programming language? Discuss in detail how C++ differs with procedural languages. 12

4. (a) Define copy constructor. What are the important features of a constructor function? 6
- (b) Using the concept of function overloading, write a program in C++ to add two numbers. 6

5. What is virtual function? With an example, demonstrate the steps for creating virtual functions.
6. What is operator overloading? Write a program in C++ program to overload prefix and postfix decrement operator by using friend function.
7. What is the use of templates? Write a C++ program to implement template to find minimum of two data items of type int, char and float.

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(2)

BCA 3rd Semester Exam., 2020

OBJECT-ORIENTED PROGRAMMING
USING C++

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer any six questions : 2×6=12

- (a) What is data abstraction?
- (b) Define object and class.
- (c) What is overloading?
- (d) Define member function.
- (e) What do you mean by static variable?
- (f) Define operator overloading.
- (g) Define exception.
- (h) What is the difference between C and C++?
- (i) What do you mean by binding?
- (j) What is thread?

AK-21/394

(Turn Over)

2. Answer any three of the following : 4×3=12

- (a) Explain how to write inline functions.
- (b) What are constructor types and its purpose?
- (c) What are super-, base, parent and inherited classes?
- (d) How to make a class member friend of another class?
- (e) Why are virtual functions needed?

3. Discuss the method overloading with a proper example. https://www.akubihar.com 12

4. List out the characteristics of procedure-oriented programming and object-oriented programming. 12

5. Discuss the uses of objects as function arguments with example. 12

6. Discuss virtual base class with an example. 12

7. List out and explain unformatted I/O operations. 12

AK-21-1900/394

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BCA 3rd Semester Exam., 2021

OBJECT-ORIENTED PROGRAMMING
USING C++

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. **1** and **2** are compulsory.

1. Answer any six questions : $2 \times 6 = 12$

- ~~(a)~~ How to define polymorphism?
- (b) Define pointers.
- (c) How to use delete operator?
- (d) What is increment and decrement operator?
- ~~(e)~~ How constructor differs from member functions?
- (f) What is base class?

(2)

- ~~(g)~~ What is pure virtual function?
- (h) What is the use of flush() function?
- ~~(i)~~ What is overloading?
- (j) What is function?

2. Answer any three of the following : $4 \times 3 = 12$

- (a) Explain function prototyping with an example.
- ~~(b)~~ Differentiate between procedure-oriented and object-oriented programming.
- ~~(c)~~ Explain inline functions.
- (d) What is a friend function? Why is it required?
- ~~(e)~~ Explain the virtual destructor.

~~3.~~ Explain different types of inheritance with block diagram and an example for each. 12

~~4.~~ What is the benefit of copy constructor? Explain the necessity of defining your own copy constructor. 12

5. Demonstrate with C++ program for (i) passing objects to functions (ii) returning object. 12
6. Explain error handling and manipulators in C++? 12
7. Explain different file operations? 12

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Code : 303301

BCA 3rd Semester Exam., 2022

**OBJECT-ORIENTED PROGRAMMING
USING C++**

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.*
- (ii) There are **SEVEN** questions in this paper.*
- (iii) Attempt **FIVE** questions in all.*
- (iv) Question Nos. **1** and **2** are compulsory.*

1. Answer any six of the following questions :

2×6=12

- (a) What is abstract class?*
- (b) What is the difference between seekg and tellg functions?*
- (c) What is the difference between setw() and setfill()?*
- (d) What is the difference between ios::ate and ios::out?*

AK23/1026

(Turn Over)

- (e) What is container class?
- (f) What is the use of extractor?
- (g) What is the purpose of 'this' pointer?
- (h) What is dynamic binding?
- (i) What is message passing?
- (j) Why is data hiding required?

2. Answer any *three* of the following questions :

4×3=12

- (a) Differentiate between object-oriented programming and procedure-oriented programming.
- (b) Explain friend function with example.
- (c) What is constructor? Explain the types of constructor with example.
- (d) Explain function overloading with example.
- (e) Differentiate between polymorphism and virtual base classes.

3. Discuss public, private and protected access specifiers and show their visibility when they are inherited as public, private and protected.

12

4. List and explain in brief various functions required for random access file operations. 12
5. What is a user-defined exception? Write down the scenario where we require user-defined exceptions. 12
6. Discuss virtual functions and dynamic binding with examples. 12
7. Specify a class called complex to represent complex numbers. Overload +, - and * operators when working on the object of this class. 12

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BCA 3rd Semester Exam., 2024

**OBJECT-ORIENTED PROGRAMMING
USING C++**

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.*
- (ii) There are **SEVEN** questions in this paper.*
- (iii) Attempt **FIVE** questions in all.*
- (iv) Question Nos. 1 and 2 are compulsory.*

1. Answer any six of the following questions :

2×6=12

- (a) Define manipulator.*
- (b) Define overriding.*
- (c) Explain generic classes.*
- (d) What is data hiding?*
- (e) What is destructor?*
- (f) What is abstract class?*
- (g) Define data security.*

- (h) Explain virtual destructors.
- (i) Explain cin and cout functions.
- (j) Define garbage collection.

2. Answer any *three* of the following : 4×3=12

- (a) What is the role of friend function in C++? Justify your answer with example.
- (b) What do you mean by encapsulation and how is it implemented in C++?
- (c) Differentiate between call by value and call by reference.
- (d) How can the* ambiguity in multiple inheritance be resolved?
- (e) Explain different types of inheritance with the help of a sample program.

3. Write a program to show the working of constructors along in inherited classes. 12

4. Write a program to show the use of friend functions. 12

5. Write a program in C++ to compute the average marks of 50 students in the class. Take necessary assumptions. 12
6. In what ways object-oriented paradigm is better than structured programming paradigm? Explain the features of OOPs. 12
7. What do you mean by inheritance? Write a program in C++ that showing the use of single inheritance. 12



Code : 303301

(2)

BCA 3rd Semester Exam., 2025

**OBJECT-ORIENTED PROGRAMMING
USING C++**

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer any six of the following questions :

2×6=12

- (a) What is the class and objects?
- (b) Define access specifier.
- (c) Explain pointer.
- (d) What is ambiguity?
- (e) Explain new and delete.
- (f) Explain friend function and friend class.

- (g) Explain token.
- (h) Explain multilevel inheritance.
- (i) Define proxy classes.
- (j) What is runtime error?

2. Answer any *three* of the following questions :

4×3=12

- (a) Write a sample code to show the difference between C and C++.
- (b) Explain the concepts of abstract classes and virtual base classes with suitable examples.
- (c) How do you define member function outside the class?
- (d) What do you mean by exception handling?
- (e) Write a program to overload "+" operator.

3. What is the significance of static data and member functions in C++?

12

4. What is a user-defined exception? Write down the scenario where we require user-defined exceptions.

12

26AK/464

(Turn Over)

26AK/464

(Continued)

(3)

5. Write a C++ program using function template to find the product of two integers or floating-point types of data. 12
6. Discuss the concepts of generalization and aggregation. 12
7. What is the role of manipulators in C++? Write down different manipulators in C++. 12
